

Chapter 1

Directional Patterns

After stabilizing 3000 trials of 301×301 sandpile grids (each with background 3, plus 3 extra grains at random sites), we examine the pattern formed by cells whose height differs from 3. Far from the center, along a fixed rational direction (n, m) , a periodic height pattern emerges. We classify these patterns for all directions with $0 \leq n, m \leq 7$.

1.1 Classification Table for (n, m) , $0 \leq n, m \leq 7$

Table 1.1 shows the directional pattern for each pair (n, m) . Only seven distinct primitive (coprime) directions were observed across all 3000 trials. When a pair (n, m) reduces by $\gcd(n, m)$ to one of these seven primitive directions, the corresponding image is shown; otherwise the entry reads “not found.”

Table 1.1: Directional pattern classification for $0 \leq n, m \leq 7$. Rows index n , columns index m . Entries show the image of the reduced coprime direction.

$n \downarrow / m \rightarrow$	0	1	2	3	4	5	6	7
0	—							
1					not found	not found	not found	not found
2				not found		not found		not found
3			not found		not found	not found		not found
4		not found		not found		not found	not found	not found
5		not found	not found	not found	not found		not found	not found
6		not found			not found	not found		not found
7		not found	not found	not found	not found	not found	not found	

The seven observed directions are:

$$(0, 1), (1, 0), (1, 1), (1, 2), (1, 3), (2, 1), (3, 1).$$

Note that (n, m) and (m, n) yield distinct patterns in general (e.g., $(1, 2)$ vs. $(2, 1)$ are different), and the negative directions $(-n, m)$, $(n, -m)$, $(-n, -m)$ produce the same patterns by symmetry. You can check the folder `images/` for the original images.

1.2 Observed Regularities

Beyond the classification, several quantitative regularities were noted across all trials.

1.2.1 Internal Color Periodicity

Along any directional line, the color pattern of heights repeats periodically without variation. For instance, a line with $(2, 1)$ always follows the sequence (brown, white, brown) from left to right.

1.2.2 Frequency of Occurrence

Lines whose slope is close to horizontal or vertical appear less frequently across trials. The vast majority of observed lines are horizontal, vertical, or at a 45° angle (slope ± 1). Lines with slopes $1/2$, $1/3$, 2 , or 3 are comparatively rare.

1.2.3 Vector Sum at Intersections

When multiple directional lines meet at a single intersection point, the sum of their unit direction vectors equals zero. This suggests a form of local conservation governing the pattern geometry.